BC FREE FLIGHT

Weather Station Field Handbook

Remote weather & camera monitoring site

Hardware · Wiring · Network · Streaming · Credentials

bcfreeflight.com/sites/coopers

Printed reference — keep on site

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> **ABOUT THIS HANDBOOK**
> Compiled from the WeatherStation repository documentation. Section 12 is intentionally blank — fill it in by hand on site. Treat the completed book as a sensitive document.

1 · System Overview

This handbook documents a remote weather station with camera monitoring. The system connects to a WiFi network and streams both real-time weather data and a live video feed of launch activities for paragliding and other free flight.

The installation uses **two ECOWITT weather stations** — one at launch and one at the landing zone — which allows the atmospheric lapse rate between the two elevations to be computed.

Live results: **https://bcfreeflight.com/sites/coopers**

1.1 System Architecture

- Solar power generation and battery storage
- Two ECOWITT weather stations (launch and LZ) with WiFi connectivity
- IP camera for visual monitoring
- Long-range wireless bridge for data transmission
- *Stream* script for live streaming to YouTube

1.2 Signal Path

STAGE	COMPONENT	CARRIES
Sense	ECOWITT GW1101 7-in-1 array (×2 — launch + LZ)	Temp, humidity, wind, rain, UV, solar
See	Reolink 12MP PoE camera	RTSP video
Power	200W panel → MPPT controller → 12V 100Ah LiFePO4	DC to all site gear
Link	PowerBeam AC Gen2 pair, bridge mode	Transparent L2 to the internet side
Serve	Mele Quieter4C mini PC + Stream script	RTSP → YouTube Live
Publish	Weather Underground / bcfreeflight.com	Weather data from both stations

1.3 Documentation Map

- **Hardware** — Parts List (§2), Wiring Diagram (§3)
- **Software & Configuration** — Software Requirements (§4), Weather Stations (§5), PowerBeam (§6), Reolink (§7), Streaming (§8)
- **Stream script source** — github.com/BCFreeFlight/Stream

1.4 Support

For questions or assistance with this setup, refer to the individual component sections for troubleshooting first. If you are still stuck, get in touch by any of the following:

CHANNEL	DETAIL
GitHub issue	github.com/BCFreeFlight/WeatherStation/issues
Streaming software	github.com/BCFreeFlight/Stream/issues
Email	**chase.florell@gmail.com**
Phone	**587-439-4852**

> **ON SITE**
> A GitHub issue is preferred for anything that is not urgent — it leaves a record others can find. Phone or email if the station is down and the fix cannot wait.

2 · Parts List

Comprehensive list of components required for building a remote weather station. There will likely be some miscellaneous components needed to complete the build, but this list is a good starting point.

> **COST**
> Total price fluctuates, but estimated cost is around **$2,400 CAD** for the launch site build. The second ECOWITT GW1101 at the LZ is an additional unit on top of that.
> Complete Amazon shopping list: amazon.ca/hz/wishlist/ls/2OBZLOWASUJUO

2.1 Compute & Networking

Mele Quieter4C Fanless Mini PC

amazon.ca/dp/B0CP3DC75C

- Intel N100 processor (12th Gen, up to 3.4GHz, 4 cores/4 threads)
- 8GB LPDDR4x RAM and 128GB eMMC storage with M.2 2280 expansion slot
- Supports 4K triple display via dual HDMI ports and USB-C
- Built-in WiFi 5 (2.4/5G) and Bluetooth 5.1 connectivity
- Compact, fanless design for silent operation and energy efficiency

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Ubiquiti PowerBeam AC Gen2 (×2)

amazon.ca/dp/B073WV2QL2

- High-performance 5 GHz wireless bridge with 25 dBi gain antenna
- Supports throughput speeds up to 450+ Mbps
- Long-range connectivity for point-to-point links up to 25+ km
- Gigabit Ethernet port for high-speed network connection
- Built-in mechanical tilt with ±20° adjustment for optimal alignment
- Improved surge protection for outdoor deployment
- Low power consumption (8.5W maximum)
- Includes 24V, 0.5A Gigabit PoE adapter for simplified power delivery

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Ubiquiti UniFi UAP-AC-M Access Point

amazon.ca/dp/B01NAAWOGO · 2.4GHz / 5GHz PoE

- Dual-band Wi-Fi access point with speeds up to 1167 Mbps (300 Mbps on 2.4GHz, 867 Mbps on 5GHz)
- Weather-resistant design suitable for outdoor installation
- Power over Ethernet (PoE) compatible for flexible deployment
- Mesh networking capability for extended coverage
- Supports up to 250+ concurrent client connections

Mini Industrial 3 Port Gigabit PoE Switch

amazon.ca/dp/B0BVH7M9M2

- 3 IEEE 802.3af/at compliant PoE ports with 10/100/1000Mbps speeds
- DC12V-48V to DC48V voltage booster for flexible power input
- Up to 30W per port power delivery for PoE devices
- Industrial-grade hardened design for harsh environments
- Din-Rail mount compatible for secure installation
- Store-and-forward switching architecture
- 4K MAC address table for efficient network management
- Supports jumbo frames up to 9K bytes for enhanced performance

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LINOVISION Industrial Gigabit 30W PoE Splitter

amazon.ca/dp/B09BQZ5ZX3

- Gigabit Ethernet support with 10/100/1000Mbps data transmission
- Switchable DC12V/24V output for powering non-PoE devices
- Passive PoE output for wireless bridges and access points
- Industrial-grade design with −40°C to 75°C operating temperature range
- 6KV lightning protection and Level 3 ESD protection
- IEEE 802.3af/at standard compliant
- Rugged metal housing for durability in harsh environments
- Wall-mountable design with included bracket for flexible installation

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2.2 Sensing & Imaging

ECOWITT GW1101 Wi-Fi Weather Station (×2)

amazon.ca/dp/B07R1Y72G4

- Solar-powered 7-in-1 outdoor sensor array for comprehensive weather monitoring
- Measures temperature, humidity, wind speed/direction, rainfall, UV, and solar radiation
- Wi-Fi connectivity for remote data access via free mobile app (iOS & Android)
- Built-in temperature, humidity, and atmospheric pressure sensors in base station
- Data upload capability to Weather Underground and other weather services
- Expandable system supporting additional sensors (soil moisture, CO2, PM2.5, etc.)
- Dual power options: solar panel and backup AA batteries for the outdoor unit
- Easy installation with included mounting hardware

> **TWO UNITS IN THIS SYSTEM**
> One at launch and one at the LZ (Landing Zone). The second unit allows for accurate computation of atmospheric lapse rates between the two locations. Each unit is registered separately with Weather Underground and has its own Station ID and Station Key.

REOLINK 12MP PoE IP Camera

amazon.ca/dp/B08P48HDVF

- Ultra HD 12-megapixel resolution for exceptional detail
- Power over Ethernet (PoE) for simplified installation with single cable
- Smart detection for people, vehicles, and pets with motion alerts
- 120° field of view provides broad coverage area
- Color night vision with 700lm illumination for clear footage in darkness
- Two-way audio communication capability
- IP66 weatherproof rating for reliable outdoor operation
- Supports up to 256GB local storage

2.3 Power

200W Monocrystalline Solar Panel Set

amazon.ca/dp/B0CZXJKQGW

- 200-watt output high-efficiency monocrystalline solar panel
- 20–22% conversion efficiency for maximum power generation
- 12BB construction for improved performance in low-light conditions
- IP65 junction box for weather resistance and durability
- Compatible with 12V battery systems
- Ideal power source for remote weather station operation
- Provides sufficient power for all electronic equipment when paired with battery storage

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15A MPPT Solar Charge Controller

amazon.ca/GCSOAR-Controller-Adapter-Control-Charger/dp/B0CXXY2VNF

- 15A maximum charging current for efficient solar power management
- MPPT (Maximum Power Point Tracking) technology for up to 30% higher efficiency than PWM controllers
- Compatible with 12V/24V systems with automatic voltage detection
- Built-in WiFi connectivity for remote monitoring and control via smartphone app
- Real-time data tracking and performance analytics accessible from anywhere
- Built-in protection against overcharging, over-discharging, and short circuits
- LED indicators for easy monitoring of system status
- Suitable for various battery types including lithium and lead-acid
- Compact design for installation in weather-resistant enclosures

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LiFePO4 12V 100Ah Lithium Battery

amazon.ca/LiFePO4-Lithium-Rechargeable-Max-15000-Trolling/dp/B0F1T7ZZPB

- 12V 100Ah deep cycle rechargeable lithium iron phosphate battery
- Built-in 100A Battery Management System (BMS) for safety protection
- Up to 4000+ charge cycles at 100% depth of discharge
- Lightweight design compared to traditional lead-acid batteries
- Consistent power output throughout discharge cycle
- No maintenance required and zero toxic emissions
- Ideal for solar power storage in remote weather station applications
- Compatible with the EPEVER solar charge controller

2.4 Enclosure, Mounting & Cabling

Junction Box — IP67 Waterproof Plastic Enclosure

amazon.ca/dp/B08282SQPT

- Dimensions: 290×190×140mm (11.4"×7.5"×5.5")
- IP67 waterproof rating for complete protection against dust and water immersion
- Durable ABS plastic construction for outdoor durability
- Hinged grey cover design for easy access to components
- Includes mounting plate and wall bracket for flexible installation
- Ideal for housing and protecting electronic components in harsh weather conditions

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Pole Mounting Kits for Junction Box

amazon.ca/dp/B0CQR42XHV

- Stainless steel bracket with hardware for outdoor installation
- 10-inch size for optimal mounting flexibility
- Adjustable clamp to fit various pole diameters
- Corrosion-resistant material for long-term outdoor durability
- Compatible with electrical enclosures and junction boxes
- Weather-resistant design for harsh environmental conditions

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Newpowa Universal Solar Panel Heavy Duty Double Arm Pole Mount

amazon.ca/dp/B0CC5TBM7Y

- Heavy-duty double arm design for secure solar panel installation
- Universal mounting system compatible with most solar panel sizes
- Adjustable tilt angle from 0–90 degrees for optimal sun exposure
- Pole/wall mounting options for flexible installation
- Durable aluminum and stainless steel components resist corrosion
- Designed to withstand high wind conditions

Cat 6 Ethernet Cable — 75 ft

amazon.ca/dp/B01F4XBTH0

- 75-foot Cat 6 Ethernet cable for long-distance networking connections
- Used at the access point end to connect the antenna to a hardwired router
- Length may vary depending on specific installation location and requirements
- Weather-resistant for outdoor installation; UV-resistant jacket
- High-speed data transmission up to 1 Gbps
- Shielded design for protection against interference in harsh environments

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Cat 6 Ethernet Cable — 10 ft (5 Pack)

amazon.ca/dp/B09ZHRWPJ7

- High-performance Cat 6 network cables with RJ45 connectors
- Supports network speeds up to 10 Gbps with 250 MHz bandwidth
- Gold-plated contacts for reliable, corrosion-free connections
- Snagless connectors to prevent clip damage during installation
- Pack of 5 cables provides sufficient connectivity for all components
- UTP (Unshielded Twisted Pair) design for standard networking applications

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Ambuker Extension Cable

amazon.ca/Ambuker-Extension-Weatherproof-Connectors-Adapter/dp/B0D83F8MGJ

- Weatherproof power extension cable for outdoor applications
- Male and female connectors with secure waterproof adapters
- UV-resistant cable jacket for long-term outdoor durability
- Ideal for connecting solar panels to charge controllers
- Provides flexible installation options for optimal solar panel placement

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Wire Loom Split Tubing

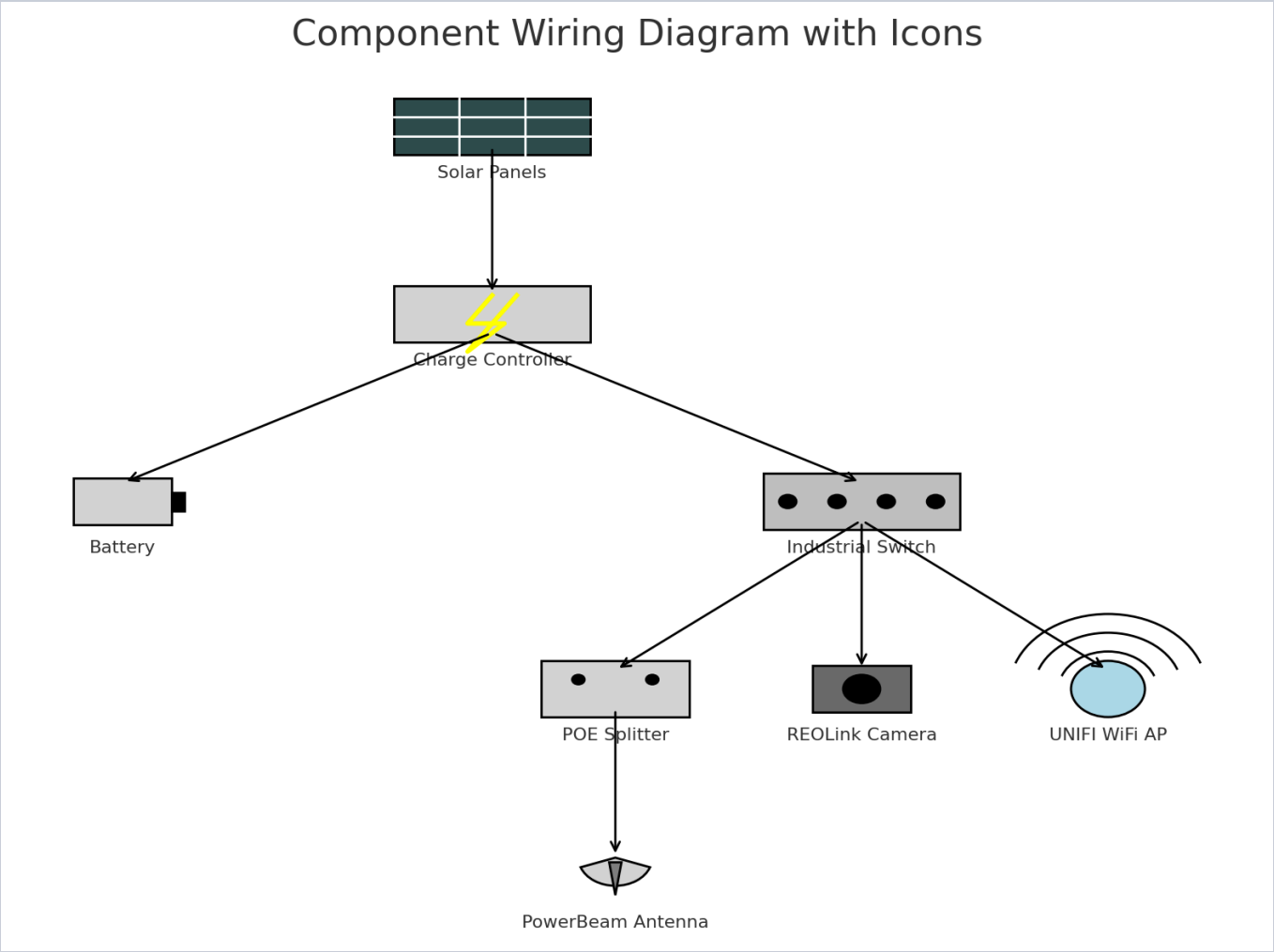
amazon.ca/dp/B09C1VQ96T

- Flexible corrugated design for easy cable organization and protection
- Pre-split construction for simple installation without disconnecting wires
- Wide temperature resistance from −40°F to 257°F (−40°C to 125°C)
- UV-resistant material for outdoor applications
- Resistant to automotive fluids, chemicals, and abrasion
- Provides protection against vibration, water, and environmental elements

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3 · Wiring Diagram

Power and network interconnect for all site components.



4 · Software Requirements

Software required to configure and service the station, with platform availability.

APPLICATION	PURPOSE	PLATFORMS
Reolink	Camera setup, live view, credentials	Android, iOS
UISP	PowerBeam discovery and management	Android, iOS
UniFi	UAP-AC-M access point management	Android, iOS
WSView Plus	ECOWITT station Wi-Fi and upload settings (both stations)	Android, iOS

> **STREAMING HOST**
> On the Mele mini PC the only software required is **STREAM**, which installs its own dependencies (`ffmpeg` and its Python packages). See §8.

4.1 Download Locations

- **Reolink** — Play Store: `com.mcu.reolink` · App Store id995927563
- **UISP** — Play Store: `com.ubnt.umobile` · App Store id1183022489
- **UniFi** — Play Store: `com.ubnt.easyunifi` · App Store id1057750338
- **WSView Plus** — Play Store: `com.ost.wsautool` · App Store id1581353359
- **Stream** — github.com/BCFreeFlight/Stream/releases/latest

5 · ECOWITT Weather Station Configuration

Feeding data to Weather Underground. Follow these steps to set up the ECOWITT weather stations and send live weather data.

5.1 Two Stations in This System

This installation uses **two ECOWITT GW1101 stations**:

STATION	LOCATION	PURPOSE
Launch	Launch site, alongside the camera and network gear	Primary launch conditions
LZ	Landing zone	Landing conditions, and the second data point used to compute the atmospheric lapse rate between the two elevations

Each station is a fully independent unit. Repeat every step below **once per station** — each one gets its own Weather Underground device registration, its own Station ID and Station Key, and its own entry in the WSView Plus app.

> **DO NOT SHARE A STATION ID**
> Using one Station ID for both units interleaves readings from two different elevations into a single feed, which corrupts both the launch data and the lapse rate. Record the elevation of each station accurately when you register it — the lapse rate calculation depends on it.

5.2 Register Each Station on Weather Underground

1. Visit Weather Underground's registration page: `wunderground.com/member/devices`
2. Log in or create an account if you do not have one.
3. Click **Add a New Device**.
4. Complete the required information, including location, elevation, and station details.
5. After registration, note the **Station ID** and **Station Key**/Password.
6. Repeat for the second station, using its own name, location, and elevation.

> **RECORD IT**
> Write both sets of Station ID and Station Key into §11 — Credentials Record. There is a separate block for the launch station and the LZ station.

5.3 Connect Each ECOWITT Console to Wi-Fi

1. Plug in the ECOWITT weather station and turn it on.

2. On the console, navigate to the network settings to enable Wi-Fi.
3. Connect your phone to the same Wi-Fi network you want the station to use.

5.4 Use the WSView Plus App

1. Download and install the WSView Plus app (Android or iOS).
2. Open the app and search for the weather station device.
3. Follow the app's instructions to connect the station to your Wi-Fi.

5.5 Configure Weather Underground Uploads

1. Within the WSView Plus app, select the weather station device.
2. Go to **Device Settings** → **Weather Services** or **Upload Settings**.
3. Select **Weather Underground**.
4. Enter the **Station ID** and **Station Key** belonging to *that* station (from §5.2).
5. Save the settings.
6. Repeat for the second station, taking care to enter the matching credentials.

5.6 Confirm Data Upload

- Return to Weather Underground's Device Management page.
- Both stations should show as "Online" after a few minutes. Data updates typically start within 10–15 minutes.
- If data does not appear, double-check the Station ID, Key, and network connection.
- If one station reports the other's conditions, the Station IDs have been swapped between the two units — correct them in WSView Plus.

> **TROUBLESHOOTING**
> Consult the ECOWITT user manual or the WSView Plus app help section.

6 · PowerBeam Bridge Configuration

Configure Ubiquiti PowerBeam antennas to communicate in bridge mode, enabling transparent network traffic between two or more locations.

6.1 Prerequisites

- Two PowerBeam (or compatible) antennas
- Devices powered up and connected via Ethernet for initial configuration
- Access to the web interface for each unit
- Default login credentials (usually `ubnt` / `ubnt`, unless changed)

6.2 Step 1 — Initial Setup

1. Connect your computer to the PowerBeam's PoE injector LAN port using an Ethernet cable.
2. Set your computer's network IP to a static address in the **192.168.1.x** range (e.g. 192.168.1.10).
3. Open a web browser and go to `http://192.168.1.20` (the default PowerBeam IP).

6.3 Step 2 — Login and Update Firmware

1. Log in (default username and password are usually `ubnt`).
2. If needed, update to the latest firmware from Ubiquiti's official site: `ui.com/download`

6.4 Step 3 — Configure the First PowerBeam (Access Point)

1. On the main menu, go to **Wireless**:
   - Set *Wireless Mode* to **Access Point**.
   - Enter an *SSID* (network name) that both antennas will use.
   - Choose your *Channel Width* and *Frequency* (try automatic, or select manually for best performance).
   - Set/encrypt with a strong WPA2 password.
2. Go to **Network**:
   - Set *Network Mode* to **Bridge**.
   - Assign a static IP if desired (to avoid IP conflicts later).
3. Apply changes and reboot if prompted.

6.5 Step 4 — Configure the Second PowerBeam (Station/Client)

1. Connect to the second antenna and log in as above.
2. Go to **Wireless**:

- Set *Wireless Mode* to **Station**.
- Enter the *SSID* to match the Access Point.
- Set Channel Width and Frequency to match the AP.
- Enter the same WPA2 password.

3. Go to **Network**:
   - Set *Network Mode* to **Bridge**.
   - Assign a different static IP, or use DHCP if your network supports it.
4. Apply changes and reboot if prompted.

6.6 Step 5 — Align Antennas and Test Connection

1. Physically position and align the antennas with each other for optimal signal (use the alignment tools in the web interface).
2. Connect your main network to the AP side and the remote device/network to the Station side.
3. Test connectivity — devices at each end should communicate seamlessly as if on the same LAN.

6.7 Tips

- Use the **UISP** app (Android/iOS) for discovery and management.
- Use surge protectors to safeguard hardware outdoors.
- Adjust frequencies to minimize interference, especially where other wireless networks are present.

For advanced settings and troubleshooting, consult your model's user manual or visit the Ubiquiti Community Forums at `community.ui.com`.

7 · Reolink Camera Configuration

Setting up the Reolink IP PoE camera and obtaining the RTSP and RTMP stream URLs. The RTSP URL is what you supply to the Stream script during installation — see §8.

7.1 Camera Hardware & Initial Setup

1. **Mount the Camera** — install at the desired location using the provided hardware.
2. **Connect Power & Network** — use a PoE adapter or switch to provide both power and network connectivity over a single Ethernet cable.
3. **Find the Camera on Your Network** — use the Reolink app (Windows/Mac/mobile) or your router's client list to note the camera's IP address.

7.2 Accessing the Camera Web Interface

1. Open a web browser.
2. Enter `http://<IP address>` (replace with the camera's assigned address).
3. Log in using your camera credentials (set up initially via the Reolink app or web interface).

7.3 RTSP Stream

RTSP is a common protocol for streaming high-quality video with minimal latency. This is the URL the Stream script uses as its `stream.rtspUrl` config value.

URL FORMAT

```
rtsp://<username>:<password>@<IP address>/Preview_<channel number>_<stream type>
```

- `<username>` — camera login username
- `<password>` — camera login password
- `<IP address>` — camera's local IP address (e.g. 192.168.1.101)
- `<channel number>` — typically `0` for single-channel cameras
- `<stream type>` — `main` for high quality, `sub` for lower quality/bitrate

EXAMPLE

Username `admin`, password `MySecret123`, IP `192.168.1.101`, main stream:

```
rtsp://admin:MySecret123@192.168.1.101/Preview_0_main
```

7.4 RTMP Stream

RTMP is typically used for pushing streams to servers (YouTube, Twitch), but Reolink cameras also expose RTMP URLs that can be pulled by a client. RTSP is preferred for this installation; RTMP is documented here as a fallback.

URL FORMAT

```
rtmp://<IP address>/bcs/channel<channel id>_main.bcs?channel=<channel id>&stream=0&user=
<username>&password=<password>
```

- `<IP address>` — camera's IP address
- `<channel id>` — usually `0` unless you have multiple channels
- `<username>` / `<password>` — camera login credentials

EXAMPLE

```
rtmp://192.168.1.101/bcs/channel0_main.bcs?
channel=0&stream=0&user=admin&password=MySecret123
```

7.5 Troubleshooting

- **Firewall issues** — ensure port 554 (RTSP) and port 1935 (RTMP) are allowed through the network firewall.
- **Multiple cameras** — repeat the above for each camera, adjusting the channel number.
- **Network stability** — a wired Ethernet connection is recommended for best reliability.

Additional resources: `support.reolink.com`

> **REMINDER**
> The RTSP/RTMP URLs embed the camera username and password in plain text. Do not post them publicly. Record the camera credentials in §11 rather than writing full URLs elsewhere.

8 · Live Streaming with the Stream Script

Live streaming to YouTube is handled by **BC Free Flight Stream** — a self-contained Python script written for this project. It proxies the RTSP camera feed to YouTube Live via `ffmpeg`, manages the YouTube broadcast lifecycle, handles OAuth 2.0 authentication, retries automatically on failure, and registers its own cron jobs for unattended operation.

Source and releases: `github.com/BCFreeFlight/Stream`

> **STABLE BROADCAST URL**
> Once created during `--install`, the same YouTube URL is reused every time the stream starts and stops. That is why it is safe to embed on the website — it does not need updating each day.

8.1 Requirements

- Linux Mint / Ubuntu-based system (the Mele mini PC at the site)
- Python 3.8 or later
- `ffmpeg` — installed automatically by `--install`
- A Google Cloud project with the **YouTube Data API v3** enabled
- An RTSP-capable camera reachable from the host — see §7

All Python dependencies are self-installed by the script on first run.

8.2 Google Cloud Setup

Do this before running the installer.

1. Go to the Google Cloud Console: `console.cloud.google.com`
2. Create a new project, or select an existing one.
3. Navigate to **APIs & Services → Library**.
4. Search for **YouTube Data API v3** and click **Enable**.
5. Navigate to **APIs & Services → Credentials**.
6. Click **Create Credentials → OAuth client ID**.
7. Select **Desktop app** as the application type.
8. Copy the **Client ID** and **Client Secret** — both are needed during `--install`. Record them in §11.

> **NO STREAM KEY TO COPY**
> You do *not* need to fetch a stream key from YouTube Studio. The installer creates the broadcast and live-stream resources through the API and writes the stream URL and key into `config.toml` automatically.

8.3 Install

On the streaming machine, `cd` to the directory where the script should live and run:

```
curl -fLO https://github.com/BCFreeFlight/Stream/releases/latest/download/stream.py \
  && python3 stream.py --install
```

The interactive wizard will:

- Prompt for Google OAuth credentials, the RTSP URL, broadcast title/privacy/category, and the cron schedule
- Install `ffmpeg` and the required Python packages
- Open a browser for Google OAuth 2.0 authorization
- Create the YouTube broadcast and live-stream resources
- Write `config.toml` and `.env` beside the script
- Detect the terminal emulator and register cron jobs
- Print the stable YouTube URL for the stream

Re-running `--install` is idempotent. It loads existing configuration and only prompts for values that are missing, preserving credentials, YouTube resources, and cron entries.

8.4 Commands

COMMAND	PURPOSE
`stream.py --start`	Create a fresh broadcast and start streaming
`stream.py --stop`	Stop `ffmpeg` gracefully and archive the broadcast as a VOD
`stream.py --recover`	Resume streaming if the current time is inside the daily window (used at boot)
`stream.py --install`	Run or re-run the setup wizard
`stream.py --update`	Back up and upgrade to the latest release
`stream.py --roll-back [version]`	Restore a previous version from `backup/`
`stream.py --reinstall`	Clean slate: uninstall, delete config, re-install
`stream.py --uninstall`	Stop the stream, archive the broadcast, remove cron entries
`stream.py --set-property <key> <value>`	Set a `config.toml` value from the command line
`stream.py --start --log-level debug`	Override log verbosity for a single run

All commands are run as `python3 stream.py …` from the directory containing the script.

8.5 Scheduling

`--install` registers three cron entries by default, scheduled for the April–October flying season, plus an optional fourth:

JOB	DEFAULT SCHEDULE	BEHAVIOUR
Start	`30 6 1-31 4-10 *` (6:30 AM)	Opens a terminal window and starts streaming
Stop	`25 18 1-31 4-10 *` (6:25 PM)	Runs headless to stop the stream
Recover	`@reboot`	Runs `--recover`; resumes the stream if the machine reboots inside the window
Update *(optional)*	`0 0 * * *` (midnight)	Runs `--update`; only registered when `cron.autoUpdate = true`

Auto-update is disabled by default so the site is never updated unattended without consent. The start job opens a single terminal window titled "BC Free Flight Stream", which closes when the stop job fires — this prevents terminal windows accumulating over the season.

8.6 Configuration

Non-secret configuration lives in `config.toml` beside the script. Frequently adjusted keys:

KEY	DEFAULT	DESCRIPTION
`stream.rtspUrl`	*prompted*	Full RTSP URL of the camera
`stream.mute`	`false`	If `true`, audio is stripped
`youtube.broadcastTitle`	`My Location: {date}`	Title template; `{date}` resolves to the current ISO date
`youtube.privacy`	`public`	`public`, `unlisted`, or `private`
`youtube.embeddable`	`true`	Allow the broadcast to be embedded on external sites
`youtube.enableDvr`	`false`	Allow viewers to rewind the live stream
`youtube.archivePrivacy`	`private`	Privacy applied to the archived video after `--stop`
`logLevel`	`info`	`debug`, `info`, `warning`, or `error`
`logRetentionDays`	`15`	Days of logs to keep
`cron.start` / `cron.stop`	see §8.5	Daily start and stop schedules
`cron.autoUpdate`	`false`	Register the automatic update job

> **SECRETS**
> Secrets live in `.env` beside the script: `GOOGLE_CLIENT_SECRET`, `GOOGLE_REFRESH_TOKEN`, `GOOGLE_ACCESS_TOKEN`. Never share or commit this file. The RTSP URL in `config.toml` also contains the camera password in plain text.

8.7 Logs

- **Location** — `logs/` beside the script, one file per day
- **Format** — `[ISO-8601-timestamp] [LEVEL] message`
- Old files are pruned on each `--start` according to `logRetentionDays`
- At the default `info` level `ffmpeg` progress noise is hidden; use `--log-level debug` to see it

8.8 Troubleshooting

Run any command with `--log-level debug` for full detail.

STREAM NEVER GOES LIVE / STATUS STUCK INACTIVE

- Verify the camera URL and that the host can reach it; check the `[ffmpeg]` debug lines.
- If the camera password changed, re-run `--install` — special characters are encoded automatically.
- A stream key mismatch is repaired by re-running `--install`.

"PLAYBACK ON OTHER WEBSITES HAS BEEN DISABLED" (EMBED BLOCKED)

- Ensure `youtube.embeddable = true` and run `--start` again.

OAUTH FAILS OR AUTHORIZES THE WRONG ACCOUNT

- The flow always shows the account chooser — pick the correct account.
- To switch accounts permanently, use `--reinstall`.

STREAM DIDN'T RESUME AFTER A REBOOT

- `--recover` only acts inside the `cron.start` / `cron.stop` window.
- Confirm the entry exists: `crontab -l | grep bcfreeflight_stream`
- Re-run `--install` to repair cron entries.

NO TERMINAL WINDOW OPENS

- The detected terminal may be wrong: `python3 stream.py --set-property terminal xterm`

PIP INSTALL FAILS (EXTERNALLY-MANAGED ENVIRONMENT)

- PEP 668 systems fall back to `--user --break-system-packages` automatically. If it still fails, pre-install the dependencies or run inside a virtualenv.

For anything not covered here, see the Stream troubleshooting guide at `github.com/BCFreeFlight/Stream/blob/main/docs/troubleshooting.md`

8.9 Credentials Involved

- Google account owning the YouTube channel
- Google Cloud OAuth client ID and client secret
- Reolink camera username/password (embedded in the RTSP URL)

Record all of these in §11 — Credentials Record.

9 · Appendix A — SubmitWeather API

BC Free Flight runs a Lambda-backed ingest endpoint for weather data.

ENDPOINT

```
https://api.bcfreeflight.com/SubmitWeather?uploadKey={KEY}
```

where `{KEY}` is the **deviceKey** associated with your device.

> **FORMAT**
> Currently only Weather Underground querystring-formatted data is supported.

Record the device upload key in §11 — Credentials Record.

10 · Appendix B — Adding a Station to BCFreeFlight.com

To get a weather station added to bcfreeflight.com:

1. Set up the weather station to send data to Weather Underground (§5).
2. Submit an issue to `github.com/BCFreeFlight/bcfreeflight.github.io/issues/new` with the following information:
   - Weather Underground Station ID
   - Weather Underground Station Key
   - Station name and location
   - Site elevation in metres
   - Brief description of the flying site
   - Contact information (optional)

The BC Free Flight team will handle the integration and add the weather station data to the site once the submission is approved.

11 · Credentials Record

HANDLE AS CONFIDENTIAL
Once filled in, this section contains live credentials for the site. Store the printed book in a controlled location, do not photograph these pages, and shred superseded copies. Cross out and re-write entries when a password is rotated — note the date.

Fill in by hand. Leave unused entries blank.

11.1 Network Infrastructure

Site Wi-Fi / SSID
Network the ECOWITT console and site gear join

SSID _____

PASSWORD _____

Router / Gateway Admin (AP end)
Hardwired router at the access point end of the bridge

IP ADDRESS _____

USERNAME _____

PASSWORD _____

PowerBeam AC Gen2 — Access Point side
Web UI, default 192.168.1.20 · default login ubnt/ubnt

IP ADDRESS _____

USERNAME _____

PASSWORD _____

PowerBeam AC Gen2 — Station side

Web UI, remote/site end of the bridge

IP ADDRESS _____

USERNAME _____

PASSWORD _____

PowerBeam Bridge Link (wireless)

SSID and WPA2 key shared by both antennas

SSID _____

WPA2 KEY _____

UniFi UAP-AC-M Access Point

UniFi app / controller

CONTROLLER _____

USERNAME _____

PASSWORD _____

Ubiquiti Account (UISP / UI.com)

Cloud account for device management and firmware

USERNAME _____

PASSWORD _____

11.2 Site Devices

Reolink Camera — Device Login

Camera web UI at http://<IP> · also used in the RTSP/RTMP URL

IP ADDRESS _____

USERNAME _____

PASSWORD _____

Reolink Cloud Account

Reolink mobile app sign-in

USERNAME _____

PASSWORD _____

Mele Quieter4C Mini PC — OS Login

Streaming host; local account and any remote-access method

HOSTNAME/IP _____

USERNAME _____

PASSWORD _____

MPPT Solar Charge Controller — Wi-Fi App

Remote monitoring app for the 15A MPPT controller

DEVICE / AP _____

USERNAME _____

PASSWORD _____

11.3 Weather Services

ECOWITT / WSView Plus Account

Account used to configure both GW1101 consoles

USERNAME _____

PASSWORD _____

ECOWITT GW1101 — Launch Console

Device Wi-Fi/network details for the launch-site station

WI-FI SSID _____

IP ADDRESS _____

ELEVATION (M) _____

ECOWITT GW1101 — LZ Console

Device Wi-Fi/network details for the landing-zone station

WI-FI SSID _____

IP ADDRESS _____

ELEVATION (M) _____

Weather Underground — Account

wunderground.com/member/devices

USERNAME _____

PASSWORD _____

Weather Underground — Station (Launch)

Station credentials entered into WSView Plus

STATION ID _____

STATION KEY _____

Weather Underground — Station (LZ)

Second GW1101 at the landing zone, for lapse-rate computation

STATION ID _____

STATION KEY _____

BC Free Flight SubmitWeather API

api.bcfreeflight.com/SubmitWeather?uploadKey={KEY}

DEVICE NAME _____

UPLOAD KEY _____

11.4 Streaming

Google Account (YouTube channel owner)

Owns the live broadcast; also used for OAuth consent

USERNAME _____

PASSWORD _____

2FA / RECOVERY _____

Google Cloud OAuth Client (Stream script)

Desktop-app credentials from console.cloud.google.com, entered at `--install`

PROJECT NAME _____

CLIENT ID _____

CLIENT SECRET _____

YouTube Broadcast

Stable URL created by `--install`; stream key is written to config.toml automatically

CHANNEL URL _____

BROADCAST ID _____

11.5 Project Accounts

GitHub (BCFreeFlight org)

Issue submission and Stream script releases

USERNAME _____

PASSWORD / PAT _____

AWS Console (SubmitWeather Lambda)

Only if you administer the ingest endpoint

ACCOUNT / ALIAS _____

USERNAME _____

PASSWORD _____

11.6 Additional Entries

Component / service name:

NAME _____

USERNAME _____

PASSWORD _____

Component / service name:

NAME _____

USERNAME _____

PASSWORD _____

Component / service name:

NAME _____

USERNAME _____

PASSWORD _____

Component / service name:

NAME _____

USERNAME _____

PASSWORD _____

11.7 Password Change Log

DATE	COMPONENT	CHANGED BY